

A Web-Based Interface for Information Propagation Analysis on Directed Acyclic Networks

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Abstract—

I. INTRODUCTION

Network analysis and graph theory algorithms have always played a central role in reliability engineering and risk assessment. From fault tree analysis to probabilistic inference, these methods are used across many areas including infrastructure planning, performance assessment, and maintenance [?], [?], [?]. Representing complex infrastructure processes as directed acyclic graphs (or classical network models) allows engineers to trace information flow, identify bottlenecks, and quantify vulnerabilities throughout a complete system lifecycle, from design through to operation.

The mathematical foundations for these approaches are well established and widely implemented in the form of different algorithms and software libraries. Unfortunately, the people who need these tools most, decision makers, infrastructure planners, and reliability engineers, are not always the same people who develop the algorithms. This is not a new problem. Graphical user interfaces have long addressed this gap in other disciplines. From statistical analysis tools like SPSS and JMP to circuit simulation with SPICE, GUIs democratize access to specialized computational methods for domain experts who are not necessarily programmers. Studies consistently report this same gap between algorithmic availability and practical adoption in the context of scientific computing and network analysis [?], [?].

In today's data-driven environment, several approaches have attempted to bridge this gap. Web-based platforms such as Jupyter notebooks [?], Shiny applications [?], and Streamlit dashboards [?] provide interactive interfaces and improved workflow sharing. However, these platforms remain fundamentally code-driven, requiring users to write Python, R, or similar scripting languages to define models and run analyses. For practitioners without programming backgrounds, this continues to limit adoption. A similar approach is taken by libraries such as NDLib [?] and CDLib [?], which offer dedicated interfaces for simulating diffusion processes and identifying community structures in networks, respectively. These are valuable for researchers but still require users to write code.

Visualization tools such as Gephi [?] and Cytoscape [?] have gained widespread adoption, but focus primarily on network topology and layout rather than probabilistic analysis. To perform a full analysis, users must combine these with specialized

tools such as GeNIe [?], BayesiaLab [?], or Netica [?], each with its own learning curve and data format requirements. What practitioners actually need is a single workflow that serves both algorithm developers and domain experts who are not necessarily programmers.

Probabilistic analysis of complex networks underpins most reliability engineering and risk assessment methods. However, existing low- and no-code software tools often restrict uncertainty representation to single probability distributions or interval bounds. In networks where data are sparse, incomplete, or derived from expert judgement, uncertainty cannot always be adequately captured by a single distribution. Probability bounds analysis (PBA) and probability boxes (p-boxes) address this limitation by representing uncertainty as bounds on cumulative distributions rather than imposing precise distributional assumptions [?]. Software libraries supporting p-box arithmetic exist for R [?], Julia [?], and Python [?], but using these methods still requires writing code directly against the libraries.

Deployment is also a practical consideration. Although cloud-based solutions offer convenience and scalability, they raise privacy and data sovereignty concerns [?]. For industries working with proprietary infrastructure models or regulated data, such as nuclear, defence, or critical infrastructure sectors, local deployment is often a contractual or regulatory requirement rather than a preference [?].

To date, no existing tool combines (i) purely visual, no-code interaction, (ii) comprehensive network analysis capabilities including structural, probabilistic, and performance analysis, (iii) native support for uncertainty quantification using p-boxes, and (iv) guaranteed local-first deployment. This paper presents a web-based graphical interface for information propagation analysis on directed acyclic networks that addresses this gap. The toolkit provides a fully no-code environment supporting multiple analysis modes: network structure exploration, network decomposition, and probability propagation with interval uncertainty. Built on the `InformationPropagationAnalysis.jl` Julia package [?], the system exposes these algorithms through a REST backend server connected to an Angular-based frontend. The system operates entirely on the user's local machine, ensuring that sensitive network data never leaves the user's environment. The decoupled architecture allows domain experts to use the interface without programming, while developers can extend

each component independently.

The remainder of this paper is organized as follows. Section II describes the system architecture, detailing both the Julia backend server and Angular frontend components. Section III presents the supported analysis modes and their corresponding interface elements. Section IV demonstrates the system through representative use cases. Finally, Section V concludes with a discussion of limitations and directions for future work.